

Indoor360

Precision Navigation Redefined for the Digital Twin Era

Presented by Kassem Aletawi

The Indoor **Blind Spot**

Modern GPS fails exactly when we need it most: inside the complex architectures of hospitals, airports, and mega-malls.

A wide margin of error leads to floor-level confusion and navigation failures, making precise destination finding impossible in dynamic indoor spaces.



The Indoor360 Solution



Floor-Aware

Responsive multi-level awareness that transitions seamlessly through elevators and hallways.



Fused Data

Synergy of IMDF files, SLAM techniques, and A* pathfinding for meter-level accuracy.



Stable Sync

Maintains an uninterrupted connection to the facility's digital twin even in the deep interior.

Unmatched Precision

~1m

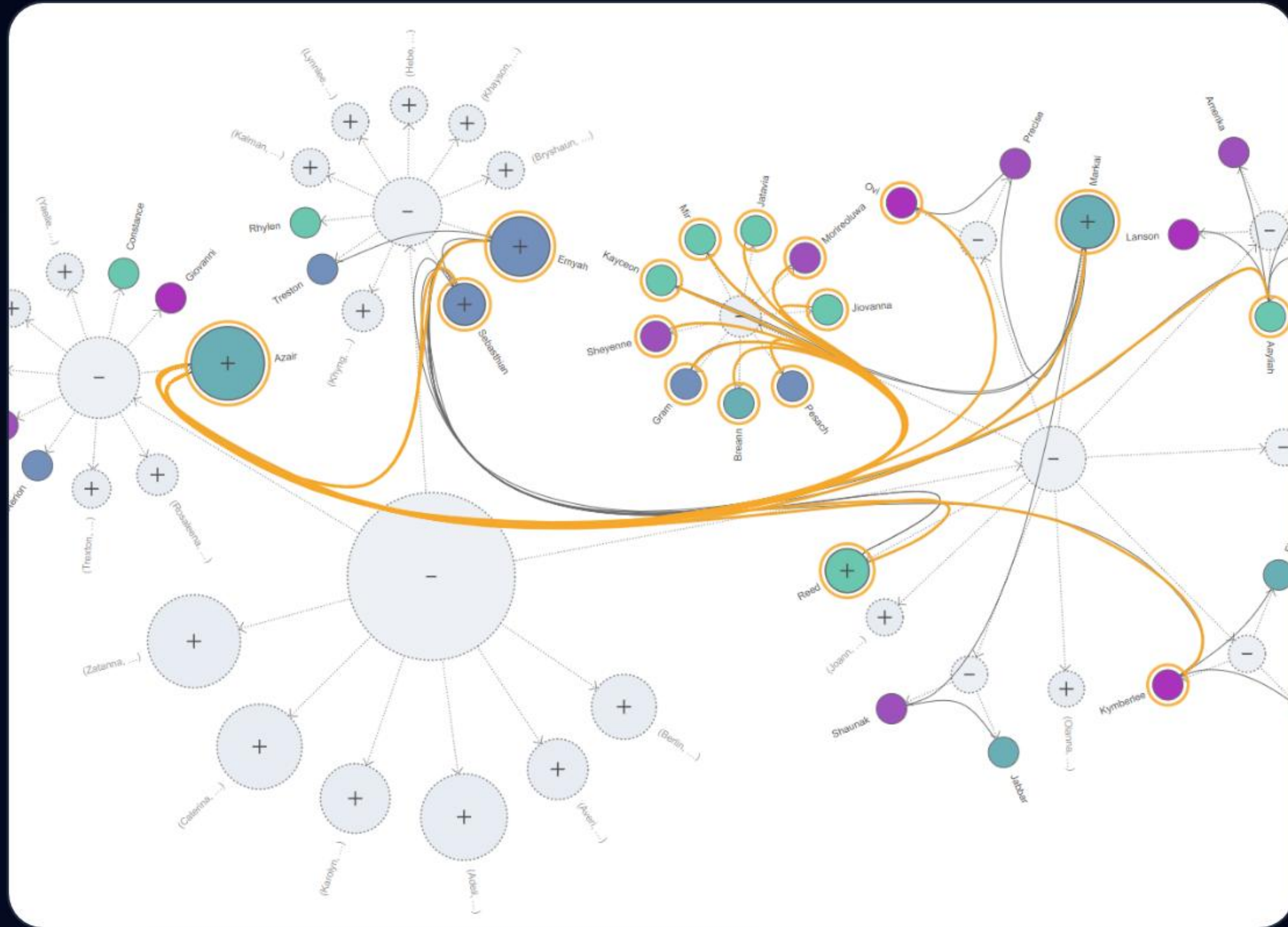
Meter-Level Accuracy

Beyond Traditional Limits

By moving beyond a single data source, Indoor360 eliminates the "blind spot" and distinguishes between specific rooms and floor levels with surgical precision.

This level of accuracy is critical for emergency services, logistics, and high-traffic public venues.

Advanced Algorithms



Optimal Pathfinding

Indoor360 utilizes the **A* Algorithm** to calculate the most efficient routes in real-time, accounting for walls, sections, and obstacles.

Simultaneous Localization and Mapping (SLAM) ensures the user's digital position stays perfectly synchronized with the physical environment.

IMDF Pipeline

From Blueprint to App

Our architecture transforms raw architectural data into actionable insights:

-  **QGIS Processing:** Geo-referencing floor maps.
-  **PostGIS Integration:** Robust spatial data handling.
-  **JSON Extraction:** Scaling IMDF features to Swift.



Premium UX/UI

Native iOS Performance

Built with Swift, the frontend offers a high-fidelity experience optimized for the modern professional.

- 🌙 **Native Dark Mode:** Energy-efficient design.
- ⚡ **Quick Actions:** One-tap routing to cafes or exits.
- 📍 **AR Support:** Future-proof guidance.



Inclusive Navigation



Step-Free Guidance

Navigation shouldn't be a barrier. Indoor360 prioritizes accessibility by calculating routes specifically for mobility-impaired users.

Our algorithm identifies ramps, elevators, and wide corridors to ensure everyone reaches their destination safely and efficiently.

Technical Stack

Layer	Technologies	Purpose
Frontend	Swift / CoreLocation	User Interface & Device Positioning
Backend	Python / FastAPI	Route Calculation & API Endpoints
Database	PostgreSQL / PostGIS	Spatial Queries & IMDF Storage
Mapping	QGIS / GeoJSON	Floor Plan Processing & Georeferencing

Algorithm Performance



Comparative analysis of indoor positioning methods demonstrating Indoor360's industry-leading precision.

Seamless Transitions

Eliminating the boundary between the physical and digital world,
one floor at a time.



Questions?

Thank you for your attention.

Indoor360 | Kassem Aletawi

Image Sources



<https://media.istockphoto.com/id/1460962791/photo/busy-airport-terminal-at-sunset.jpg?s=612x612&w=0&k=20&c=oG8RY1MoZy9J4EejeSjyNhwGPpQvuADIR-twzpalItg=>

Source: www.istockphoto.com



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